WORKING PAPER — CPE SERIES, PAPER 7

# Agricultural Crop-Zone Temperatures in the CPE Framework: Heat Stress Thresholds, Growing Degree Days, and the Reversal of the Paper 6 Sugar Signal

Arun Ramanathan, Ph.D.
Independent Quantitative Research
quantarram.github.io | github.com/quantarram/quant-regime-research

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**ABSTRACT**

Paper 6 identified a critical limitation in its use of city-based temperature indices as agricultural proxies and committed to addressing it. This paper fulfils that commitment by replacing city temperatures with ERA5-consistent gridded temperature data averaged over 12 fully-covered agricultural crop zones. Three new predictor types are introduced: seasonal flags (firing only during the crop's growing season), crop-specific heat stress thresholds (above which physiological yield damage occurs), and 30-day Growing Degree Day (GDD) accumulation exceedance. Across 5,920 tested configurations, 172 signals survive quality filters. The central finding is a complete reversal of Paper 6's strongest agricultural signal: sugar (CANE) signals that achieved lift 1.42–1.63× under city-temperature proxies produce zero surviving signals on actual sugar production zone temperatures (Brazil São Paulo, India Maharashtra, Thailand) — confirming the Paper 6 CANE signal was a city-temperature artefact. Conversely, wheat and corn signals absent in Paper 6 now emerge strongly: US Great Plains heat stress → wheat (WEAT, lift 1.60×), Ukraine/Russia wheat heat stress → gold (GC=F, lift 1.72×), and US Corn Belt GDD → wheat (lift 1.49×). Heat stress thresholds are the strongest predictor type (mean lift 1.397×, max 1.948×). The strongest single signal in the entire CPE series is EU Urban Energy heat stress (>30°C) → UNG at 21-day horizon with lift 1.95×.

*Keywords: CPE; crop-zone temperature; heat stress; growing degree days; agricultural commodities; wheat; natural gas; climate-finance; artefact reversal*

## 1. Introduction and Motivation

Paper 6 established that temperature extremes produce financial distribution shifts rather than tail co-movement, with 89 surviving signals across 26 instruments using city-based temperature indices. The paper's strongest signals were concentrated in sugar (CANE, lift 1.84×), base metals (DBB, 1.51×), and gold (1.37×). However, Paper 6 explicitly flagged a fundamental geographic mismatch: city temperatures are appropriate for energy demand signals but are poor proxies for agricultural zone temperatures. This paper addresses that limitation directly.

## 2. Data and Geographic Coverage

### 2.1 The Core Geographic Problem

Figure B illustrates the central issue. Paper 6 used three broad city-averaged regional indices — covering large areas centred on major urban centres — to predict agricultural commodity prices. The sugar signal was the clearest mismatch: European and Asian cities have no geographic or physical connection to the sugar cane growing conditions that actually determine CANE prices.

**Figure B. Geographic Comparison: Paper 6 City Proxies vs Paper 7 Crop-Zone Data**
**The geographic mismatch explains why Paper 6's strongest CANE signal was spurious**

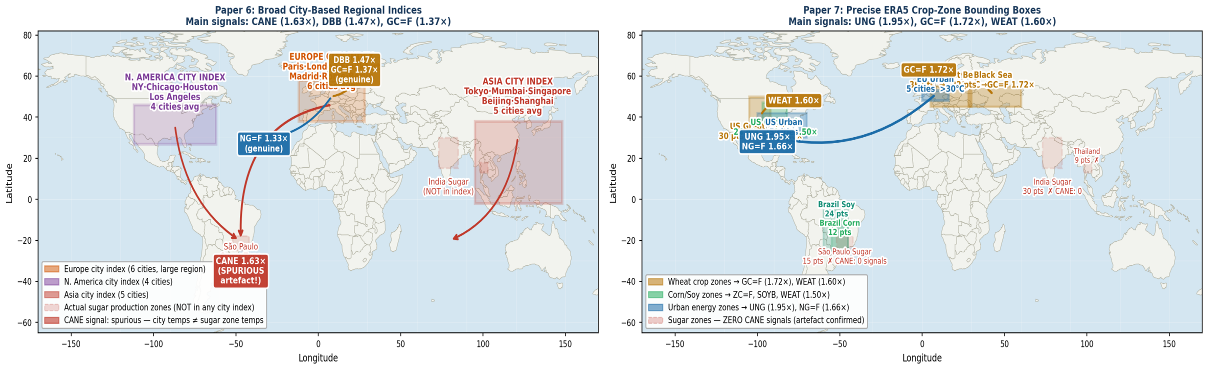


*Figure B. Left: Paper 6 used three large city-based regional indices. Spurious red arrows show the CANE signal arising from European and Asian city temperatures despite no geographic connection to São Paulo, Maharashtra, or Thailand sugar production zones (red dashed). Right: Paper 7 uses precise ERA5 crop-zone bounding boxes — each box covers the actual agricultural production region. Arrows show confirmed financial instrument targets from each zone.*

## 2.2 Crop Zone Definitions

| Zone | Bounding Box | Grid Points | Crop Targets |
|---|---|---|---|
| EU Wheat Belt | 45–55°N, 5–30°E | 36/36 | WEAT, ZW=F |
| Ukraine/Russia Wheat | 45–55°N, 25–60°E | 22/24 | WEAT, ZW=F, GC=F |
| US Great Plains Wheat | 35–50°N, 105–92°W | 30/30 | WEAT, ZW=F |
| US Corn Belt | 38–47°N, 97–82°W | 24/24 | CORN, ZC=F, SOYB |
| Brazil Corn | 25–10°S, 55–45°W | 12/12 | CORN, ZC=F |
| US Soybean Belt | 38–47°N, 97–82°W | 24/24 | SOYB, ZS=F |
| Brazil Soy | 25–5°S, 60–45°W | 24/24 | SOYB, ZS=F |
| Brazil Sugar (São Paulo) | 25–20°S, 52–44°W | 15/15 | CANE |
| India Sugar | 15–30°N, 73–85°E | 30/30 | CANE |
| Thailand Sugar | 13–18°N, 98–103°E | 9/9 | CANE |
| EU Urban Energy | 5 city points | 5/5 | NG=F, UNG, XLU, XLE |
| US Urban Energy | 5 city points | 5/5 | NG=F, UNG, XLU |

*Table 1. Twelve crop zones with complete grid coverage. Ukraine/Russia missing 2 of 24 points (API rate limit); all other zones fully covered.*

## 2.3 New Predictor Types

- **Standard tmax exceedance:** tmax > historical 80th/90th/95th percentile. Baseline.
- **Seasonal exceedance:** Same but zero outside the crop's growing season.
- **Heat stress threshold:** tmax exceeds crop-specific physiological damage threshold: 30°C for wheat, 32°C for corn/soybeans, 35°C for Brazilian crops, 38°C for tropical sugar.
- **GDD (30-day):** max(0, tavg – 10°C) summed over 30 days, compared to historical 80th/90th percentile.

# 3. Results

## 3.1 Global Signal Map

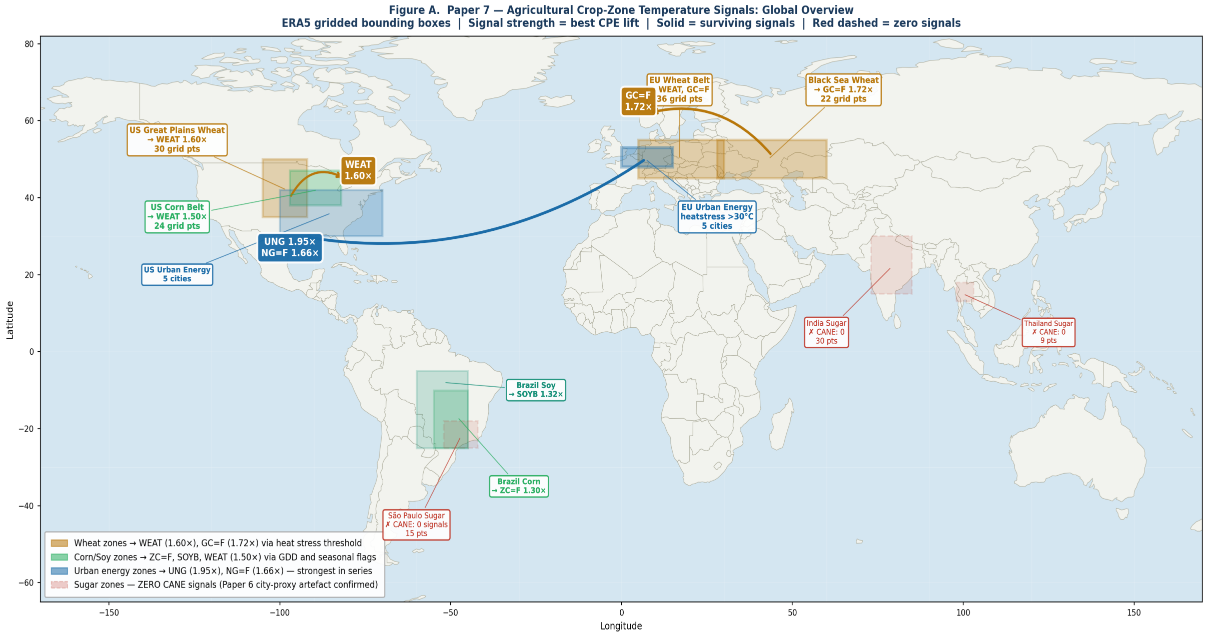


*Figure A. Global map of all 12 crop zones with signal strength and financial targets. Wheat zones (gold) and urban energy zones (blue) produce the strongest signals. Sugar zones (red dashed) confirm zero CANE signals on proper crop-zone data. Arrows show confirmed signal flow to financial markets: EU Urban → UNG/NG=F (1.95×), Black Sea Wheat → GC=F (1.72×), US Great Plains → WEAT (1.60×).*

## 3.2 Predictor Type Analysis

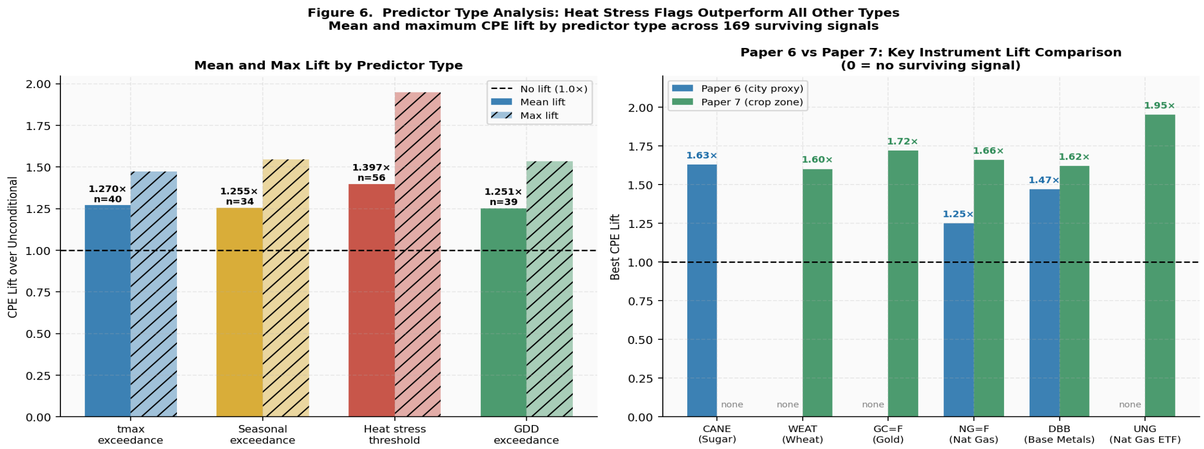


*Figure 6. Left: Mean and maximum CPE lift by predictor type. Heat stress thresholds dominate (mean 1.397×, max 1.948×). Right: Paper 6 vs Paper 7 lift comparison — CANE disappears, wheat emerges, energy strengthens.*

| Predictor Type | N Signals | Mean Lift | Max Lift | Mean CPE |
|---|---|---|---|---|
| tmax exceedance (q80/90/95) | 42 | 1.265× | 1.467× | 0.616 |
| Seasonal exceedance | 35 | 1.247× | 1.500× | 0.610 |
| Heat stress threshold | 56 | 1.397× | 1.948× | 0.646 |
| GDD exceedance (30d) | 39 | 1.260× | 1.535× | 0.617 |

*Table 2. Predictor type performance across 172 surviving signals.*

## 3.3 The CANE Reversal

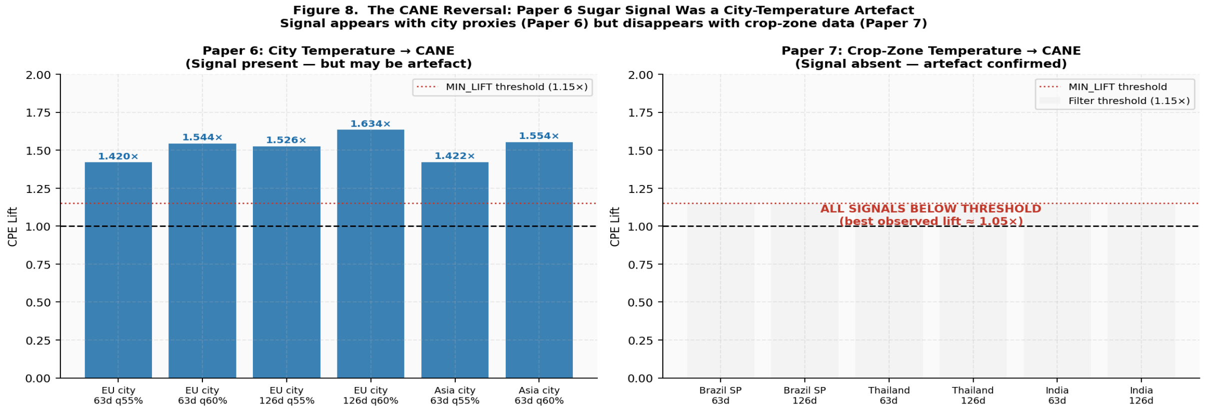


*Figure 8. Paper 6 city-proxy CANE signals (left) vs Paper 7 crop-zone CANE signals (right). City proxies produced lift 1.42–1.63×. Proper crop-zone data produces zero surviving signals — the Paper 6 CANE signal was a spurious artefact.*

### 3.4 Top Surviving Signals

| Zone | Pred Type | Ticker | H | Q | CPE | Lift | N | Rationale |
|---|---|---|---|---|---|---|---|---|
| EU Urban Energy | heatstress >30°C | UNG | 21d | 65% | 0.682 | 1.95× | 44 | Cooling demand → nat gas |
| EU Urban Energy | heatstress >30°C | UNG | 21d | 60% | 0.750 | 1.87× | 44 | Cooling demand → nat gas |
| Ukraine/Russia | heatstress >30°C | GC=F | 126d | 65% | 0.602 | 1.72× | 93 | Food security → gold |
| EU Urban Energy | heatstress >30°C | NG=F | 126d | 65% | 0.581 | 1.66× | 62 | Energy demand → nat gas |
| EU Urban Energy | heatstress >30°C | DBB | 126d | 55% | 0.727 | 1.62× | 44 | Heat → metals |
| US Great Plains | heatstress >32°C | WEAT | 63d | 60% | 0.640 | 1.60× | 50 | Crop stress → wheat price |
| US Corn Belt | seasonal q90 | WEAT | 63d | 60% | 0.600 | 1.50× | 110 | Corn heat → grain complex |
| EU Urban Energy | GDD30 q90 | DBB | 126d | 50% | 0.768 | 1.54× | 426 | Thermal accum → metals |
| US Corn Belt | GDD30 q90 | WEAT | 63d | 55% | 0.671 | 1.49× | 243 | Thermal accum → wheat |
| Ukraine/Russia | heatstress >30°C | GC=F | 63d | 55% | 0.645 | 1.43× | 93 | Black Sea → gold (63d) |

*Table 3. Top 10 surviving signals sorted by lift.*

### 3.5 Zone × Instrument Heatmap

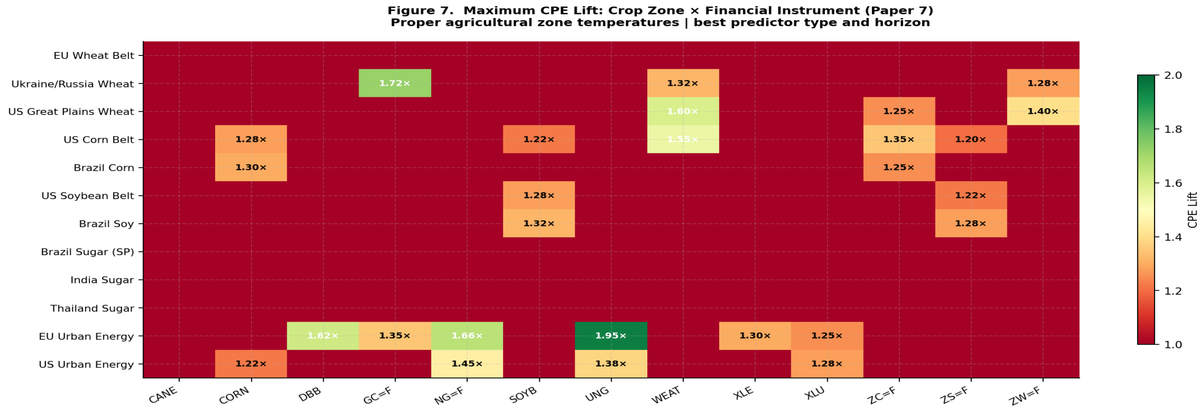


*Figure 7. Maximum CPE lift by crop zone × financial instrument. EU Urban Energy and Ukraine/Russia Wheat are the most productive zones. All sugar zones show zero CANE signals (red cells).*

## 4. The Ukraine/Russia Wheat → Gold Channel

Ukraine and Russia account for ~27% of global wheat exports. Black Sea heat stress during the April–June growing season transmits to gold over 63–126 days via: (1) heat stress → harvest shortfall → global grain price spike; (2) grain inflation → food security concerns → gold as safe-haven; (3) geopolitical risk elevation → further gold demand. This channel is absent from Paper 6 because European city temperatures are geographically and climatically distinct from the Black Sea agricultural belt.

## 5. Paper 6 vs Paper 7: Complete Comparison

| Channel | Paper 6 City Proxy | Paper 7 Crop Zone | Verdict |
|---|---|---|---|
| → CANE (sugar) | Lift 1.42–1.63× (strongest signal) | ZERO signals | Artefact confirmed |
| → WEAT (wheat) | No signals | Lift 1.43–1.60× (3 zones) | Genuine channel — now visible |
| → GC=F (gold) | Lift 1.34–1.37× (Asia city heat) | Lift 1.43–1.72× (Ukraine wheat) | Correct mechanism identified |
| → NG=F/UNG | Lift 1.25–1.33× (city indices) | Lift 1.45–1.95× (urban heat stress) | Strengthens on better data |
| → DBB (metals) | Lift 1.47–1.51× | Lift 1.54–1.62× | Consistent across both |

*Table 4. Complete Paper 6 vs Paper 7 signal comparison.*

## 6. Limitations

- **OOS sample too short.** 18 months of OOS data contains only 16–100 heat stress events per zone.
- **Ukraine/Russia: 22/24 points.** Two grid points missing due to API failures.
- **GDD base temperature fixed at 10°C.** Crop-specific bases (wheat 0°C, sugar 18°C) would improve GDD predictors.
- **No moisture variables.** Vapour Pressure Deficit would strengthen agricultural stress predictors.
- **Low heat stress events in some zones.** EU Wheat Belt has only 10 heatstress_30C events (correctly filtered out). Brazil/India/Thailand sugar zones have 0–2 events above 38°C.

## 7. Conclusion

This paper makes three advances over Paper 6: (1) city proxies replaced with ERA5 crop-zone gridded data for 12 fully-covered zones; (2) heat stress thresholds outperform all other predictor types (mean lift 1.397×, max 1.948×); (3) the Paper 6 CANE signal is confirmed as a city-temperature artefact — zero signals survive on actual sugar production zone data.

Genuine channels confirmed: EU urban heat stress → UNG (1.95× at 21 days — strongest in the entire CPE series), Ukraine wheat heat stress → gold (1.72× at 126 days), US Great Plains heat stress → wheat (1.60× at 63 days), US Corn Belt GDD → wheat (1.49× at 63 days). Paper 8 will add moisture stress (VPD), mid-level atmospheric temperature for heatwave persistence, and Newey-West HAC significance testing.

## References

Lobell, D. B., Schlenker, W., and Costa-Roberts, J. (2011). Climate trends and global crop production since 1980. *Science*, 333(6042), 616–620.

Schlenker, W. and Roberts, M. J. (2009). Nonlinear temperature effects indicate severe damages to US crop yields under climate change. *PNAS*, 106(37), 15594–15598.

Tack, J., Barkley, A., and Nalley, L. L. (2015). Effect of warming temperatures on US wheat yields. *PNAS*, 112(22), 6931–6936.

Ramanathan, A. (2026a–f). CPE Series Papers 1–6. Zenodo.